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JEA: On the Road to Cost Transparency

John B. MacArthur
University of North Florida

Jeffrey E. Michelman
University of North Florida

Bobby E. Waldrup
University of North Florida

Dana M.Wallace
Florida State University

BACKGROUND

In late spring 2006, Wanyonyi Kendrick sat in her 16th floor office of the JEA[1] tower in Jacksonville, Florida, wondering if the next step in her quest for improved cost transparency of technology services (TS) would work. As she watched the afternoon thundershowers roll in across the St. John's River, Wanyonyi, JEA's chief information officer (CIO), wondered what would happen over the next several months. JEA is the eighth largest community-owned electric, water, and wastewater utility in the United States.[2]

Wanyonyi was going over all that occurred in the last two years and contemplating what the next year would bring. She was hoping the hurricane season would not hit the north Florida coast as hard as it did in 2004. During 2004, the four major hurricanes that hit Florida caused devastation throughout the state. Although Jacksonville did not receive a direct hit from any of the hurricanes, the storms did bring mass power outages to the area. Wanyonyi also wondered what would have happened in August of 2005 if Hurricane Katrina had turned east instead of west when it hit the Gulf of Mexico. The power failures of 2004 made worse the challenges of providing customer service at JEA. Because of this, JEA's TS costs had increased significantly in an effort to prevent failures such as the JEA website going down during the 2004 hurricanes.

Unfortunately for Wanyonyi and the rest of the country, the forecasters were not giving her much hope for the 2006 hurricane season either and were predicting an increased number of intense storms. Wanyonyi had a lot to be concerned about. It was now May 5th and the hurricane season would be here in less than a month.

Not only did Wanyonyi have the weather to worry about, but there had been change within the organization as well. In the last several years the chief executive office (CEO) position had turned over. The current CEO, Jim Dickenson, placed increased emphasis on JEA's recommitment to operational excellence in the wake of the 2004 outages. His increased demands for excellence dramatically increased the strain on TS's resources. Wanyonyi had experienced trouble determining just exactly how much these new demands were increasing costs. There were gaps in the reports produced, and she had trouble putting the pieces together. It seemed she wasn't alone. At TS Department meetings, directors had voiced their complaints that the current reports seemed obsolete due to the ever-changing environment.

Wanyonyi reflected that in early 2005 she did not know how to solve this puzzle. The TS Department expenses were hidden from users and suffered from a lack of transparency. She wanted to find a way to prevent TS's current process-based costing[3] (PBC) system from failing. The current PBC system had been designed to cost activities in the context of processes and subprocesses to support JEA's process orientation,[4] but it contained a number of design flaws and provided largely obsolete information. Wanyonyi needed a PBC system that would help her establish cost savings despite the increased demands placed on the TS system to lower the risk that it would not be able to function in severe hurricane weather conditions. She had to continuously increase the level and quality of technology services to meet the CEO's strategic objectives while maintaining a focus on lowering costs. It was of strategic importance to increase the customer value of TS operations while reducing costs.

Wanyonyi was glad that she had hired three accounting professors from a well-respected university in late spring 2005 to review the usefulness of the existing PBC TS reports and to recommend improvements. Several years previously she had hired other consultants to evaluate TS's costing system, and although their work was of high quality, it did not produce actionable items.

The past year seemed to fly by and now she stared at the final report from the professors. It recommended the next steps to be taken to improve the PBC system to achieve transparency of TS costs. She felt good about her interaction with the team[5] and their final report, but she was not sure that she would be able to get buy-in throughout the organization. Wanyonyi needed a plan to get the PBC initiative integrated into the balanced scorecard process of other members of JEA's executive management.[6]

THE PBC INITIATIVE AND PROPOSAL

Because Wanyonyi was both a CPA and a CMA, it seemed that she was constantly taking continuing professional education (CPE) classes. She got a big surprise in April 2005 when she took a CPE course on strategic cost management at the well-respected university. During a coffee break she spoke to the instructor, Professor Jones, about her costing problems and he seemed quite interested. At the end of class they agreed to meet the following week at her office.

The following week, Dr. Jones, along with two of his colleagues, met Wanyonyi to discuss the problems currently facing the TS Department and the environment at JEA. During the discussion they all came to an agreement that a proposed agenda should be made and presented to Wanyonyi's boss and CEO, Jim Dickenson. Although Wanyonyi knew she could approve a project of this size out of her own budget, she needed to present the proposed benefits of this project to the CEO and get his buy-in before moving forward. If they could make this work, it could help invigorate the dying corporate PBC initiative that had been started several years earlier. She really hoped that by having TS take a leadership role, her Department could become a model for the rest of the organization.

Wanyonyi had been involved with several previous unsuccessful PBC initiatives at JEA, and the application of the adopted PBC system had also stalled. These initiatives had been unsuccessful largely because of insufficient executive management support on a continuing basis. Insufficient effort had been devoted to obtaining ongoing executive buy-in to the PBC concept, especially given the recent CEO turnover. Wanyonyi knew that the CEO's support was crucial to the success of this PBC project. The tone must be set at the top of JEA if this was to be a lasting solution. She also felt that this might be an opportunity for her area to use the COBIT Framework as a means to develop a costing model.[7] In particular, she thought that COBIT's Maturity Model[8] concept would be a way to evaluate the process. The PBC team all agreed with Wanyonyi's analysis and set a deadline for the proposal to be completed by May 16, 2005. In the meantime, Wanyonyi agreed to brief the CEO on the new initiative.

May 16th quickly came, and Dr. Jones and his colleagues arrived as planned at Wanyonyi's office. She had already read and reread the PBC proposal (see Figure 1) and was excited by the ideas. Dr. Jones seemed very calm and confident about the steps presented in the outline. Wanyonyi liked the structure of the phases and repeated interaction between the professors and her staff. The professors proposed a three-stage action plan to generate future cost savings in TS expenditures. The plan included:

1. **Transparency—**renewal of useful PBC metrics, which was essentially just showing the vice presidents (VPs) the magnitude of TS costs.
2. **Best practices—**creation of a formal chargeback system in which users of TS resources were accountable for TS expenses.
3. **Outsourcing—**if management could obtain TS resources cheaper from another vendor, they could utilize that option.

Wanyonyi confirmed her commitment to the proposal and set up points of contact between the professors and several of her directors to start Phase 1 in July. Wanyonyi breathed a sigh of relief and hoped that this would help to solve the lack of transparency and escalating costs problems.

UNDERSTANDING THE EXISTING PBC INITIATIVE[9]

The three faculty members arrived at JEA at 10 a.m. on July 5th, ready to go. They were met by Steven Schultz, Director of Technology Infrastructure and responsible for the largest portion of TS production departments. Their first appointment was with Ray Tull, a director in the Accounting Department.

Ray: Good morning.

Dr. Jones: Good morning. Thank you for taking the time to sit down with us.

Ray: No problem. The second quarter is closed, so I have a little breathing room. Why don't we make our way up to my office?

Dr. Jones: That sounds great.

Figure 1:
Process-Based Costing Proposal

Statement of Purpose

This proposed study addresses the needs of the TS Department of JEA in the area of performance measurement. The purpose of this study is to assist the TS Department of JEA in its ongoing process-based costing (PBC) deployment. The timeframe is one calendar year, beginning July 1, 2005. The study will accomplish five outcomes:

1) Assessment of the current maturity state of TS's performance metrics use, and recommendations for achieving process improvements translating into tangible cost savings.
2) Assessment of the current mix of collected performance metrics, and recommendations towards greater concentration in financial measurement.
3) Primary education of TS management as to the value and use of PBC metrics and the layered balanced scorecard system for performance measurement
4) Recommendation for the use of developed financial metrics toward directed management compensation issues.
5) Follow-up education of TS management as to how the metrics are collected and the manner in which success will be evaluated.

The first initiative of cost transparency will consist of the following six phases:

1) Researcher Shadowing and Investigating
2) Assessment of Current Processes and Metrics
  a. Qualitative evaluation of previously collected performance metrics
  b. Statistical testing of previously collected metrics
  c. Investigation of new potential quantitative metrics
3) Process Maturity Model
  a. Development of qualitative maturity-level assessment of JEA TS deployment efforts for PBC use as a decision tool
  b. Deliverable of a written maturity model, including suggestions for future growth
4) Primary Management Education
  a. Interactive Seminar with TS directors and managers
    i. Pre-test assessment of TS management's current understanding
    ii. Discussion of research findings from Phases 1-3
    iii. Interactive conversations about JEA culture, value, use of PBC
5) Financial Performance Metrics
  a. Face-to-face meeting with TS executive management
    i. Formal recommendation for financially-linked metric(s)
    ii. Discuss data collection requirements for new metric(s)
  b. Deployment of data collection techniques and time lines
6) Follow-up Management Education
  a. Final face-to-face meeting with TS executive management
    i. Finalize agreement as to outcomes assessment from new metric(s)
    ii. Agree to terms of project outcome dissemination among TS management
  b. Interactive seminar with TS directors and managers
  c. Written deliverable to TS executive management of overall study findings

As they made their way to Ray's office, Dr. Jones gave a brief summary of the questions he would be asking and the areas he would like to cover in the interview.

Dr. Jones: We really need to understand the current PBC system from the beginning. Can you walk us through the set-up of the current PBC system?

Ray: Well, we have made several attempts beginning in 1998. The current PBC system began development in 2000. JEA hired an outside consulting firm that the Accounting Department worked closely with to create the current PBC system. Most of the JEA folks who worked on it are no longer employed here. Together, we decided upon the activity cost pools, activity drivers, and the allocation rates that are used throughout JEA. Each cost pool is aligned vertically under processes and assigned to a director. Each director oversees anywhere from 30 to 50 activity cost pools. The costs of some resources can be traced to an activity cost pool, but the costs of shared resources have to be allocated over activity cost pools. Currently, the activity cost pool costs are not assigned to users.

Dr. Jones: How does JEA currently collect and distribute data?

Ray: We use an Oracle application to collect and distribute reports company-wide. The data is available to directors monthly on a shared server.

Dr. Jones: Now that we understand the setup of the current PBC system, I would like to know about the efficiency of the current Oracle system. This might be more in Steven's area, though. Would you say that Oracle requires significant resources to fully implement the data collection and reporting with the current PBC?

Steven: No. I would say that we are currently trying to determine the appropriate level of investment in personnel and technology with respect to Oracle.

Dr. Jones: Can you tell me a little more about how the activity cost pools are currently established?

Steven: Well, in 2002 we decided to drop traditional costing and implement PBC. In 2001, JEA changed from a functional organizational structure to a lean, process-oriented organizational structure.[10] PBC was adopted to support the new process organizational orientation, which helps to avoid local optimization of interdependent processes. The entire organization's processes were documented into process maps. Each cost pool now has a corresponding process map. There are hundreds of these process maps. Each process is then traced to a departmental director for accountability. For an example, I am over 35 cost pools. Does that help explain it?

Dr. Jones: Yes it does—thanks. My colleagues and I would like to see a sample of the current process flowcharts before we leave. How are the indirect costs allocated?

Ray: I think I can field that question. For indirect costs we use a standardized inter-corporate chargeback system of 11 processes. Examples include Audit, Facility Services, and Risk Management. If you want more information on how this is allocated, I can obtain that for you.

Dr. Jones: Yes, that might be helpful to have. What about the activity drivers?

Ray: Well, we refer to those as "units of measure" here at JEA. Each activity pool is assigned a corresponding unit of measure. They are not always mutually exclusive to the cost pools, but there is very little overlap.

Steven: One issue I notice is that the activity drivers are not always current or up to date. For example, the number of desk telephones is one of my activity drivers I see on the PBC report, but the number is never accurate. Other activity drivers I see on the report are obsolete. For example, no one in TS carries a pager anymore, yet it is still on my report. Come to think of it, there is very little in that report that is useful.

Ray: You are not the first director from TS that I have heard that from.

Dr. Jones: I would like to see a copy of those reports, too. Is there anything else that the PBC reports break out?

Ray: Yes, unit rates. They are mathematical derivations of the assigned dollars in the activity cost pools divided by the actual activity driver values.

Dr. Jones: OK. Thanks for taking the time today. That should be about it for now. I would just like to get a copy of those process maps and the reports.

Steven took the three professors back to Wanyonyi where they discussed what they discovered during their interview, and she filled in the missing pieces about the 2002 project. Wanyonyi also introduced them to other directors within the TS Department, and they established internal contacts that they could use for e-mail correspondence to obtain further input.

Wanyonyi: Yes, in 2002 we tried to do the whole system at once. It was really overwhelming. We only had one person on the staff and no clear executive support. The good news is we can produce numerous reports. Unfortunately, most of the information is obsolete, with very little decision usefulness. TS is viewed as a necessary evil in today's environment, but getting the resources to support it is almost impossible. Have you heard the saying, "TS is the business and the business is TS"?

Dr. Jones: No, I have not heard that before.

Wanyonyi: Basically, the saying highlights the idea that TS is intertwined in virtually all business processes today. Thus, it is difficult to determine the value of TS expenditures to show they are cost-beneficial.

Dr. Jones: That is so true, and that's why we are here to help! From the preliminary interviews, it appears that we will need to focus on understanding consumption of the services of the TS Department and how they drive costs. It also seems that the information that is currently produced does little to assist the directors in their decisions.

Wanyonyi: Yes, you hit the nail on the head.

The rest of the afternoon the professors shadowed several TS managers to better understand their processes, which are the root of the activity drivers. Over the next several months, the professors wrapped up their questioning of the TS directors and completed Phase 1 of the proposal as scheduled—researcher-shadowing and investigating. The next of the six phases, assessment of current processes and metrics, followed on their agenda.

ASSESSMENT OF CURRENT PROCESSES AND METRICS

During the next few months, Dr. Jones and his team reviewed their notes regarding the TS Department organization as well as its internal processes. Dr. Jones also received e-mails from Ray that illustrated JEA's current year data collection and classification. With all this information at their fingertips, the team of professors determined the major gaps in previous PBC initiatives at JEA, developed an action analysis of TS resource consumption, and also established the direction the project needed to take at that point. The professors met with Wanyonyi to discuss these issues.

Dr. Jones: We basically discovered five major gaps in your company's past attempts at PBC initiatives. These gaps include:

1. **Lack of decision usefulness**
   - Because the activity drivers are both dated and not related to cost behavior patterns, they are being ignored by managers and employees.
2. **Inadequate activity drivers**
   - The TS activity drivers are mainly measures of frequency—the number of times an activity occurs—which does not adequately allocate activity costs when the amount of time to perform each activity is not uniform and a time measure of intensity is needed.
3. **Outdated activity driver values**
   - The activity driver measures (called "values") are out of date and largely represent actual activity in 2001, when PBC was implemented across JEA, rather than measuring the current level of activity.
4. **Dependence of activity drivers on actual values**
   - Using actual costs can result in excessive costs from inefficiencies being unfairly transferred to customers rather than staying in the TS Department where they belong.
5. **Lack of cost management focus**
   - The existing PBC model assigns costs to processes at a macro level that is suitable for determining long-run chargeback rates, but it is less suitable for cost management efforts to better serve internal customers while reducing process costs. Effective cost management requires micro level analysis of activities within processes.

Wanyonyi: The five points are all important. So what do you suggest we do to fix these problems?

Dr. Jones: Well, I think we could start by first focusing on just one internal customer of TS resources, such as a VP or officer of a specific department. We would like to sit down with such a manager to discuss his or her department's consumption of TS resources and ultimately create a report for that manager that says, "Here is how you're consuming TS resources, and here are the activities you can change to reduce costs in your area."

Wanyonyi: I would like to tell you that getting the other VPs involved will be easy, but in fact it is going to be more difficult than I envisioned. I just left an executive staff meeting and the decision has been made to shut down Ray Tull's PBC operation at the end of December. The system has been on life support for quite some time now, but no one wanted to pull the plug. For this reason we are going to need to make a compelling business case before I can get my colleagues on the executive committee to buy into this initiative.[11]

Dr. Jones: I appreciate the dilemma, but if they saw an example of this improved report and if we showed them how they could save money in their department, maybe it would spark some interest. Our team has already established what the reports would look like. Here is an example (see Figure 2). We also believe that by using the Maturity Model framework from the outset, we may be able to provide a systematic means for implementing the costing model.

Figure 2:
Process-Based Costing Report Example[a]

FYE 2006	
Action Analysis of Technology Services Consumption[b]	
Total TS Resources Consumed by Internal Customers	$61,463,030
Green Costs	
License Fees and Maintenance	(xxx,xx)
Professional Services	(xxx,xx)
Supplies	(xxx,xx)
Labor	(xxx,xx)
Personal Devices	(xxx,xx)
Other	(xxx,xx)
Capital Investment	(xxx,xx)
Yellow Costs	
License Fees and Maintenance	(xxx,xx)
Professional Services	(xxx,xx)
Supplies	(xxx,xx)
Labor	(xxx,xx)
Personal Devices	(xxx,xx)
Other	(xxx,xx)
Capital Investment	(xxx,xx)
Red Costs	
License Fees and Maintenance	(xxx,xx)
Professional Services	(xxx,xx)
Supplies	(xxx,xx)
Labor	(xxx,xx)
Personal Devices	(xxx,xx)
Other	(xxx,xx)
Capital Investment	(xxx,xx)
Total TS Resources Consumed by XX Area of Company	$61,463,030

[a] Source: Waldrup, MacArthur, and Michelman (2009).
[b] The action analysis report with color-coding scheme was adapted from Garrison, Noreen and Brewer (2010).

Wanyonyi: OK, so what is the difference between the green, yellow, and red costs?

Dr. Jones: The green costs are variable or mixed and can be reduced directly by director-level decisions. An example of a green cost initiative would be reducing the number of cell phones or PDAs in a department. Next, the yellow costs are discretionary fixed and can be reduced by vice president-level decisions. A possible yellow cost item might be a software application, such as the asset management system, in which a department is only using one feature but is paying for three that are available. That VP could try to eliminate the unused licenses and thus realize some cost savings. Last, red costs are structurally fixed and can be reduced only by executive-level decisions. These costs are on a greater scale, such as the cost of a network.

Wanyonyi: Interesting. So we would give each VP and/or officer one of these reports, and from there he or she could identify potential cost savings, whether the savings are in the green, yellow, or red area?

Dr. Jones: Exactly. Each of the internal customers could use the report as a template for future project-related action. The VPs could also integrate these cost reports with JEA's Six Sigma initiatives aimed at continuously improving processes.[12]

Wanyonyi: I think this might work! Oh, I just had an idea.

Dr. Jones: Yes?

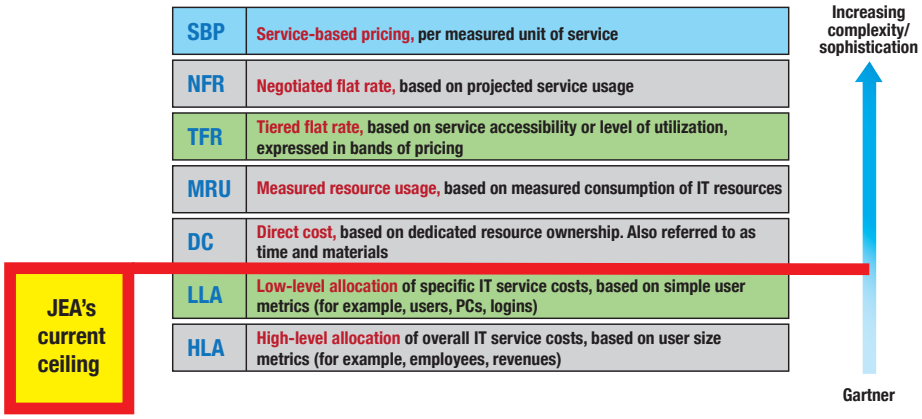
Wanyonyi: Yes. I think you need to show this all to the CEO to update him. That way, if we have his backing, he can essentially endorse the project in order to gain more VP support. Once that happens, it will be easier for you to get a pilot department VP working with you on this project. I think this is just what we need to get PBC back on track. Let's plan to meet with Jim right after the New Year. My assistant will set it up.

Dr. Jones: That would be great, but I am still concerned that we may have trouble getting VP buy-in.

Wanyonyi: I understand your concern, but I have already included a PBC metric, the number of dollars saved through cost transparency, as part of my balanced scorecard for next year. My plan will be to get my colleagues on the executive committee to apply these as well. In this way I hope we can look at TS resources and costs more strategically.

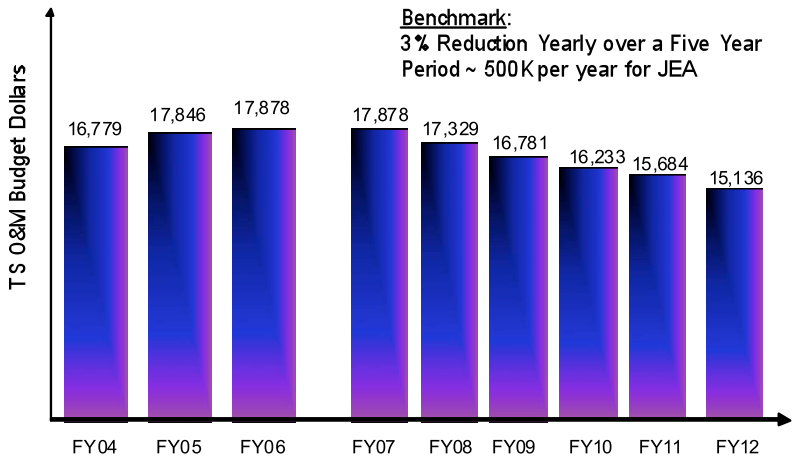
On January 5th the professors presented the five gaps, the sample VP report, and the tricolor costing scheme to update CEO Jim Dickenson on the cost transparency project's progress. Dr. Jones explained to Jim the cost transparency maturity levels, the level at which JEA was currently operating (see Figure 3), as well as the potential future savings that would potentially occur due to this cost savings initiative (see Figure 4).[13]

Figure 3:
Seven Common Methods of Service Cost Allocation[a]



[a] Adapted from Heine, 2005.

Figure 4:
Potential Future Savings from Transparency Project



Jim: I really like what I see here, Dr. Jones. You all have done an excellent job.

Dr. Jones: Thanks, Jim. It was really important to gain your support for this initiative, since we are going to need the cooperation of a key department and some of the higher-level employees, when we know they already have a lot on their plates. We also know that PBC has not engendered a great reputation around here.

Jim: I totally understand. It really is too bad that the last PBC initiative failed. I am even more committed to making this one work. You have my full support. We have to work together to reduce costs.

Dr. Jones: That's right! We look forward to getting back to you with some results.

After the meeting Steven e-mailed Dr. Jones that he and Wanyonyi had decided that the Fleet Department (Fleet) would be an excellent starting point. Although the department was small (it currently consumed $586,212 of TS's products), it was a promising unit for the initial PBC pilot study and the team would not become mired in the

details of a major production department. Moreover, Fleet was part of the Facilities and Logistics Group, which was one of TS's largest customers (it currently consumed $4,090,288 of TS's products).

THE FLEET DEPARTMENT INTERVIEW

Steven asked Dr. Jones to send an e-mail to Rachel Davis, a Fleet operations analyst, explaining the study that the team was conducting for JEA concerning cost transparency and the need for some information regarding the department. In particular, the team needed information on Fleet's organizational structure, process maps, and budget information. Rachel discussed the e-mail and scope of the study with her boss, who instructed her to e-mail Dr. Jones any information he needed. Rachel scheduled an appointment with Dr. Jones and his colleagues to meet on the following Wednesday to discuss all the information she had e-mailed him.

On Wednesday morning, March 1st, the professors and Rachel met in Steven's office. The professors briefed Rachel on the purpose of the study, discussed their roles, and then they jumped right into inquiries regarding Fleet's consumption of TS resources.

Dr. Jones: Basically we are taking everything your employees do in Fleet and mapping it to TS services. So what we have done so far is obtained your process maps, and the first thing we want to do with you is flip through them and have you tell us if they are current.

Rachel: No problem. I actually just updated most of the processes, so I can just make sure your copies are the most recent ones we have… Yes, they all look right! We have 1,464 vehicles right now. About 1,000 of those are drivable/motorized vehicles and the other 400 are trailers, excavators, and similar equipment. We have 13 employees, and this year we have a budget of $5 million.

Dr. Jones: Great. So what is your major expense in terms of TS resource consumption?

Rachel: I would have to say the software system called Maximo.[14]

Dr. Jones: Could you please explain Maximo to us?

Rachel: Sure. We use it for asset management. So every Fleet vehicle we have is in there, and even after a vehicle is sold it remains in Maximo as a retired asset (taking up storage space). Not only is Maximo used for asset management, but it's also used for getting needed work orders completed—anything from a light bulb to a whole new engine needing to be replaced. Maximo has actually been used by JEA for eight years, and Fleet just implemented it last December.

Dr. Jones: I see. Steven, when Fleet started using Maximo, what did that change in TS? Did you have to buy new licenses, modules, increase servers, or what?

Steven: Yes, this added data, a lot of data, which increased the database size, which increased the computing power that was needed to process using the Maximo software. We bought an entire new server that has eight processors in it, but sadly we are only using about half of one processor, which is only 9% utilization.

Dr. Jones: Really? Wow, there are some cost transparency issues there!

Steven: Tell me about it.

Dr. Jones: In your current budgeting system, what denominator base is used to assign TS's budgeted costs to users?

Steven: I don't know. What does that mean?

Dr. Jones: Well, in the case of Fleet requiring a new server which is only 9% utilized, is the budgeted server cost included in Fleet's budget based on its current actual usage of the new server space, the budgeted amount that Fleet is expected to use, or a practical capacity supplied rate that facilitates assigning to Fleet the budgeted costs of the spare capacity that its requirements generate?[15]

Steven: Right now, I think just the 9% of server space that Fleet currently uses.

Dr. Jones: This raises the issue of the identification of, and responsibility for, the cost of unused capacity in a major way. Speaking of cost transparency, Rachel, could you explain to us all the TS costs that appear in Fleet's budget?

Rachel explained how TS costs such as license fees and maintenance, professional services, supplies, labor, and personal devices were included in her budget. With this information, the professors were able to produce a sample report for Fleet, classifying its entire TS budget into green, yellow, and red costs. This report served as a model (e.g., see Figure 2 above) for the rollout of all reports for the other departments within JEA.

THE FINAL REPORT

The professors ultimately handed Wanyonyi the Phase 3 process maturity model final deliverable. It consisted of several key documents to help TS managers repeat the PBC analysis the professors had completed with Fleet. The professors developed a "PBC cube," which summarized the breakdown of the cost allocation system (see Figure 5). The business units were listed on the left side of the cube, the services (or applications) ran down the front of the cube, and the cost types/resources were listed on the top of the cube. The cross sections of this cube were actually mapped into a huge Microsoft Excel spreadsheet, which broke down the entire TS budget for all the JEA units. The professors set up one final meeting to discuss the report with Wanyonyi and the JEA directors involved in the project.

Wanyonyi: I understand the cube and how it relates to the Excel spreadsheet. I also understand that each business unit, or VP, will receive a report highlighting that unit's green, yellow, and red costs.

Dr. Jones: That is correct. And through use of this report, the VPs can develop cost-saving initiatives.

Wanyonyi: Right, I understand that. I am wondering, though, how this system will be maintained from year to year. I mean, the budget changes every year, not to mention the scope of certain costs also changes, so this report is really only good for one year.

Steven: That's where I can help. I was thinking that the annual budget is a good proxy of what departmental expenditures will be. So we could tie the cost allocation process to the budget process to ensure that our numbers for the reports are updated annually. I could create a database program, pull the Excel line items into the database, and essentially "force" users to update their budgets at the beginning of each year.

Dr. Jones: That sounds like a great idea. That way, it is not something that the internal users can ignore or just place on the back burner for a while. It becomes an immediate need at the beginning of each year.

Wanyonyi: Exactly. I like the idea, too.

Dr. Jones: Within this deliverable, we have also provided you with process maps (e.g., Fleet, Figure 6) that outline 1) updating the TS database (e.g., "UNIX stores data to update Maximo"), 2) gathering cost allocation details about TS customers (e.g., "Coordinator enters all required info and approves the work order"), and 3) producing the customer report (e.g., "Exception Reports").

Wanyonyi: Great. That will be helpful from year to year, and will also assist us when new leadership takes over a department. I think this initiative was a great success and will help with our cost transparency issues. Thank you.

Dr. Jones: It was our pleasure. Thank you.

Later that day, while sitting in her office and staring at the report, Wanyonyi got to thinking about the success of the cost transparency project. The end deliverable was meaningful and helpful to JEA's cause, but it was very different from what Wanyonyi had originally hoped for in reading Dr. Jones's original proposal. She recalled Dr. Jones's action plan, which consisted of:

1. **Transparency—**renewal of useful PBC metrics, essentially just showing the VPs how much TS expenses are costing them
2. **Best practices—**creation of a formal chargeback system in which users of TS resources are accountable for TS expenses
3. **Outsourcing—**if management can obtain TS resources cheaper from another vendor, they can utilize that option

Wanyonyi had read this proposal approximately one year ago to the day, and the JEA/professor team had only focused its analysis on number one—transparency—of the action plan. Scheduling meetings with department leaders had been challenging for the professors, which slowed the overall progress of the project. Wanyonyi knew the journey through numbers two and three would be difficult and would present its own challenges, but the thought that JEA would grow as an organization through realized cost savings kept her hopeful about the future of this project. How could she get her peers on the executive committee to integrate the PBC results into their respective balanced scorecards? If she could accomplish this, it would significantly improve the contribution of TS as a strategic resource to the organization.

QUESTIONS FOR DISCUSSION

1. Analyze and discuss how TS customers should use the green (short-run), yellow (medium-term), and red (long-run) cost structure for planning, control, and cost management.
2. Analyze and discuss how the cost transparency initiative can be implemented by a new computer program that is understandable to and accepted by all internal users.
3. How can JEA motivate the managers, directors, and/or VPs to change their spending behavior using PBC information?

Figure 5:
PBC Cube

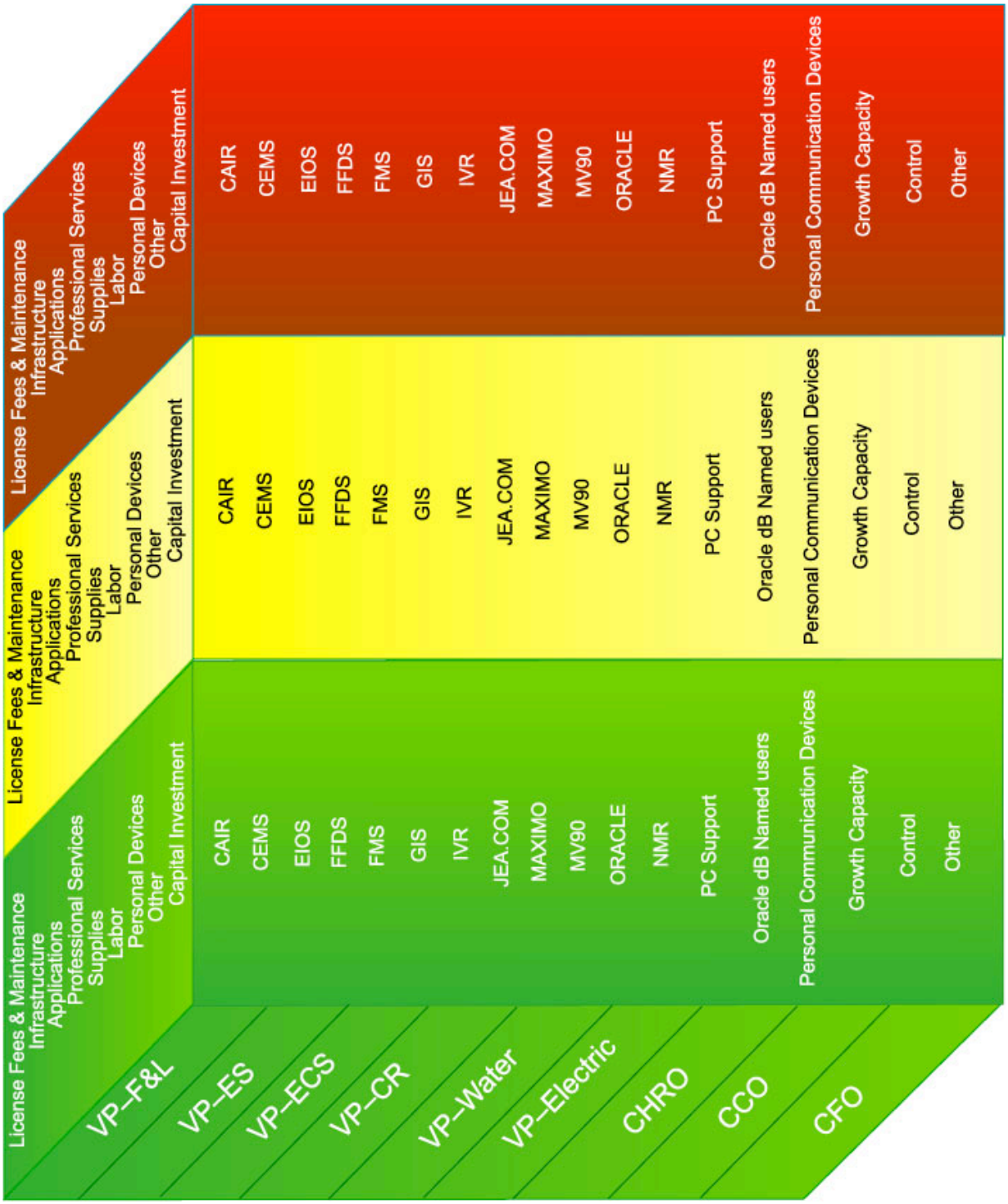
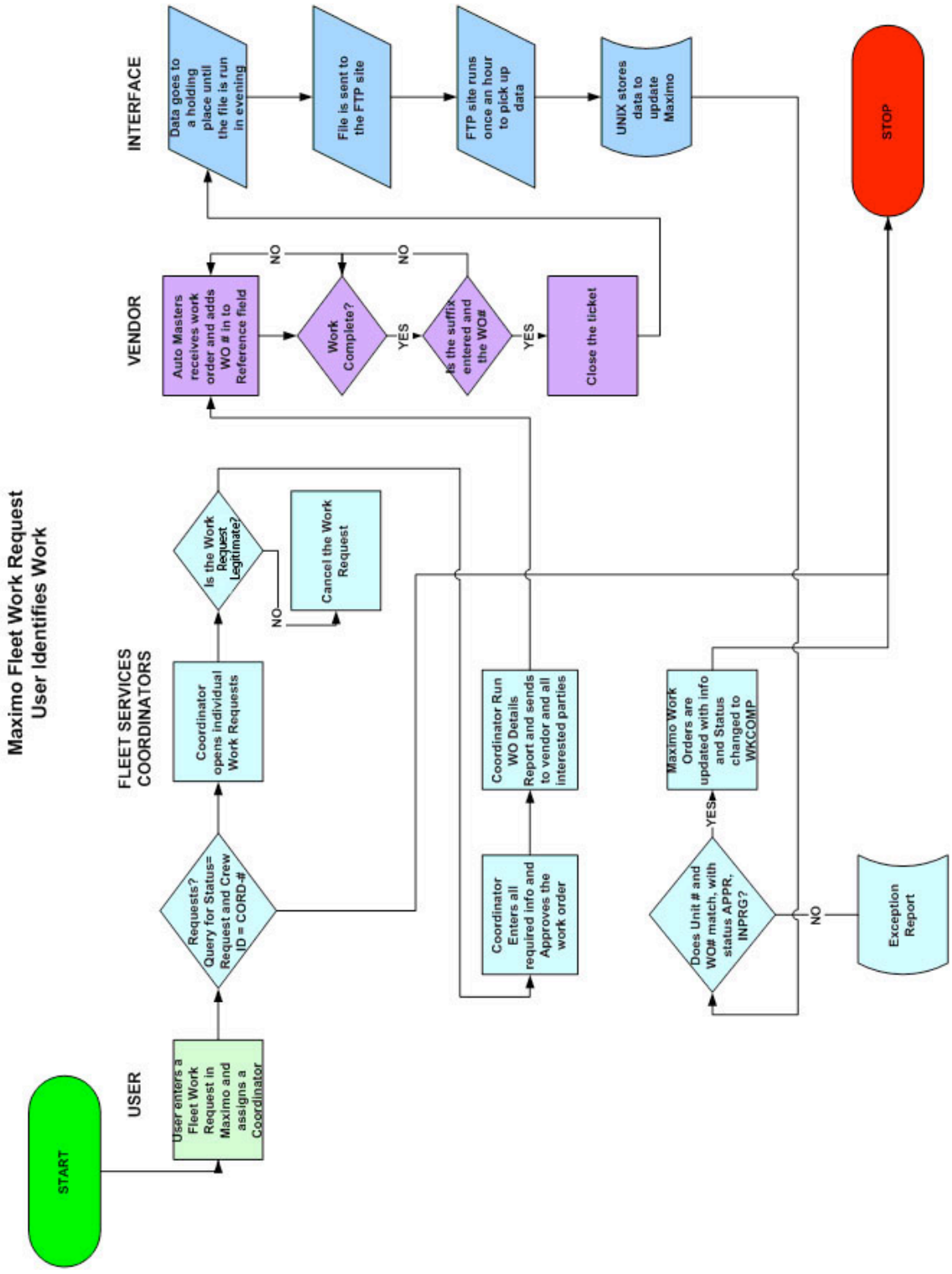


Figure 6:
Fleet Department Process Map



4. Explain the importance of having an executive champion to set the "tone at the top" for implementing strategic cost initiatives such as PBC.
5. Explain how to integrate the PBC system with JEA's Six Sigma quality improvement initiative and the balanced scorecard executive performance measurement system.

APPENDIX

Background Information on the United States Public Utilities Industry

Four different subsectors comprise the public utilities industry in the United States: Electric, Natural Gas, Telephone, and Water/Wastewater Treatment. Further, these utilities are also categorized by ownership, which is broken into three groups: Municipal, Investor, and Cooperative. In the United States there are 598 electric utilities, made up of 159 investor-owned, 251 municipally-owned, and 188 cooperatives. Water and wastewater utilities are grouped together and there are a total of 605 in 2009.[16] JEA is somewhat unique in that it is comprised of both electric and water/wastewater treatment. The electric system grew from a department of the Jacksonville city government to an independent authority created by the consolidation of city and county governments in 1967. On June 1, 1997, the water and sewer systems operated by the city since 1880 also became part of JEA's utility service offerings. Today, JEA is the largest community-owned utility in Florida and the eighth largest in the United States.[17] Florida's electric utility market is comprised of 34 municipally-owned utilities, five investor-owned utilities, and 18 rural cooperatives. An investor-owned electric utility is organized as a tax-paying business usually financed by the sale of securities in the free market, with management selected by its board, which represents shareholders. In contrast, a municipally-owned electric utility system is owned and/or operated by a municipality engaged in serving residential, commercial, and/or industrial customers, usually within the boundaries of the municipality. Finally, a cooperatively-owned utility is a joint venture organized for the purpose of supplying electric energy to a specified area. Municipal and cooperative public utilities are generally exempt from the federal income tax laws. The Rural Electrification Association has financed most cooperatives (Florida Public Service Commission, 2009, p. 1). Both electric and water/wastewater utilities are regulated by individual state public service commissions. All three types of utilities—investor, municipal, and cooperative—must request permission from their respective regulators to raise rates based on increasing costs. Investor-owned utilities must also provide evidence to support their cost of capital (Cross, 2008). In Florida, a major driver of costs in recent years has been both the increasing cost of energy (e.g., oil) and the significant cost of preparing for, and cleaning up after, hurricanes and tropical storms.

ENDNOTES

1 Additional information about JEA can be found at http://www.jea.com. (JEA is the complete name and is not an acronym.)

2 For background information on the public utility industry in the United States in general and in Florida in particular, see the Appendix.

3 For general information about PBC in theory and in practice, see Lawson (1994, 1996). In essence, PBC adopts activity-based costing (ABC) concepts but focuses on cost management of activities within the context of processes in order to optimally improve value for customers and achieve cost control. ABC systems can focus on activities in isolation that can lead to suboptimization of processes and inferior products and services for customers.

4 For further details of JEA's existing PBC system, see MacArthur et al. (2004), pp. 20-23.

5 The team was made up of three accounting professors with expertise in technology and costing systems, three graduate students in accounting, and two TS directors—one who had a significant technical background and was responsible for a significant portion of TS's operations, and the other who was responsible for the budgeting and control aspects of TS. For the purposes of this case, the three accounting professors are represented by "Dr. Jones."

6 For a discussion and examples of JEA's balanced scorecard (called Corporate Scoreboard at JEA) and a discussion of individual manager scoreboards, see MacArthur et al. (2004), pp. 15-20.

7 For information about COBIT, go to the following Internet site: www.isaca.org/cobit/.

8 COBIT draws on the Capability Maturity Model (CMM) of the Software Engineering Institute at Carnegie Mellon and focuses on developing benchmarks for continuous improvement of the five levels in the software development process (cf. Wademan, Spuches, and Doughty, 2007). Further, the CMM is particularly appropriate in this context as it considers a staged

approach to organizational change, which TS was intending to accomplish at JEA by using PBC.

9 For further details of the existing PBC system as well as its designed support for Six Sigma (called Target Smart at JEA) initiatives, see MacArthur et al. (2004), 20-23.

10 For further information, see Hightower and Bussels, 2001.

11 JEA had recently reorganized the executive leadership into the group comprised of the chief executive officer, chief operating officer, chief financial officer, chief information officer, chief human resources officer, and chief public affairs officer. The VPs of the eight operating areas then reported to this group.

12 For example, PBC supports JEA's Six Sigma initiatives by reporting resultant cost savings and by costing nonvalue-added activities (MacArthur et al., 2004, p. 20).

13 See Levinson and Pastore (2005) and Pastore (2005) for a discussion of how the Southern Company, a large electrical utility, was able to improve service and reduce information technology costs by $10 million with help from a cost transparency initiative.

14 For information about Maximo Asset Management, go to the following Internet site: www.ibm.com/software/tivoli/products/maximo-asset-mgmt/.

15 For a discussion of cost allocation based on budgeted usage, actual usage, and practical capacity supplied, see Horngren et al. (2009), pp. 545-546.

16 http://www.utilityconnection.com/index.asp

17 http://www.jea.com/about/history/index.asp.

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